

Trajectory Tracking Control for a Quadrotor Helicopter based on Backstepping using a Decoupling Quaternion Parametrization

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Abstract—This paper presents a trajectory tracking control for the position and heading angle of a quadrotor helicopter. The control design is based on the backstepping approach and proves the asymptotic stability of the tracking problem. We use a beneficial quaternion-based attitude parametrization, which is composed of two rotations. The first describes the orientation of the thrust vector independently from the second, which describes the quadrotor's heading. As a result, the translational dynamics and the yaw dynamics decouple from each other, which augments the control design. In addition, this decoupling allows an analytical derivation of all control signals within the backstepping procedure with reasonable expense. Experimental results show the practical use of the suggested control design.

I. INTRODUCTION

Quadrotor helicopters are light and very agile aerial robots and belong to the group of vertical take-off and landing vehicles (VTOL). They are useful for search and rescue missions as well as for aerial survey and inspection applications. Quadrotors are driven by four rotors symmetrically arranged at each end of a cross-shaped frame, see Fig. 1. The four single-rotor thrusts produce one total thrust for the translational acceleration and three torque components for the rotation of the vehicle. The control design for VTOLs is a challenging task and a popular benchmark example, since their dynamics are underactuated and highly nonlinear.

Trajectory tracking control for VTOLs has become a wide field of research in the last decades. While researchers initially focused on linear control techniques for slow motions near steady-state hover conditions, they have begun more and more to apply nonlinear control approaches, which allow a wider range of flight maneuvers.

In addition to other nonlinear control approaches, such as feedback linearization [1], sliding mode control [2] or forwarding [3], backstepping [4] is an often applied approach to VTOLs. Backstepping is very suitable for the cascaded plant structure of VTOLs and leads to a natural and intuitive control design. It yields a constructive way of creating a Lyapunov function for the closed loop system and thus facilitates the stability proof. A drawback is the calculation of the analytic derivatives of the virtual control signals, which is often difficult and cumbersome. In [5], the authors remedy this problem by using a sliding mode estimation, whereas in [6], command filters are suggested to generate these signals.

Several techniques can be found in literature for parametrization of the quadrotor's attitude. Euler angles are

an often used parametrization [5], but suffer from extensive calculations with trigonometric functions and singularities. A parameterization in the special orthogonal group $SO(3)$ avoids singularities and is globally unique [7], but is highly redundant and lacks in transparency. Quaternions are not globally unique, however are still useful since they allow a transparent attitude description [8], [9]. This transparency helped us to develop a decoupling attitude parametrization that we have already used in [10] for a quasi-static feedback linearization for the translational dynamics of a quadrotor. In this work we show its advantage associated with the backstepping approach.

Our contribution is summarized as follows: Due to our attitude parametrization, we can decouple the dynamics for the translational movement and the dynamics of the heading angle. This allows us to design two independent tracking control laws, one for the position and one for the heading angle, both of which guarantee the asymptotic stability of the complete system. By means of this decoupling, we are able to derive all control signals within the backstepping procedure with reasonable expense and avoid additional filtering. Moreover, we eliminate the ambiguity in our quaternions and formulate the deviation of the thrust vector by a vectorial difference between the actual and the desired thrust vector in $\mathbb{R}^3$. This enables the design of a control law that always allows rotation through the smallest angle to compensate the attitude error.

The remainder of the paper is organized as follows: Section II introduces the quadrotor model used for the control design and its quaternion-based attitude parametrization. Section III shows the backstepping control design and Section IV provides experimental results.

The superscript $'$ is used throughout this paper to indicate the transpose of a vector or matrix.

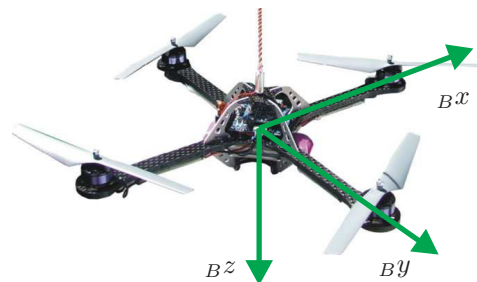


Fig. 1. Quadrotor helicopter with body-fixed coordinate frame B .

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II. QUADROTOR MODEL WITH A QUATERNION-BASED ATTITUDE PARAMETRIZATION

Before we prepare our attitude parametrization from [10] for the backstepping procedure, we provide the necessary quaternion fundamentals.

A. Fundamentals

A unit quaternion $\mathbf{q} \in \mathbb{R}^4$ with $\|\mathbf{q}\| = 1$ is a suitable parametrization of the rotation between two coordinate frames [11]. Thereby, the four elements of the quaternion

$$\mathbf{q} = [q_1 \ q_2 \ q_3 \ q_4]' = \left[\mathbf{a}' \sin \frac{\alpha}{2} \ \cos \frac{\alpha}{2} \right]' \quad (1)$$

carry the angle and axis information of this rotation, where $\mathbf{a} \in \mathbb{R}^3$ is the axis and α the angle of rotation. The transformation of a vector $\mathbf{B}\mathbf{x}$, which is given in a frame called B (denoted by the index B), to a frame called I is

$${}_I\mathbf{x} = \mathbf{R}(\mathbf{q}) \mathbf{B}\mathbf{x}, \quad (2)$$

with

$$\mathbf{R}(\mathbf{q}) = \begin{bmatrix} q_4^2 + q_1^2 - q_2^2 - q_3^2 & 2(q_1q_2 - q_3q_4) & 2(q_1q_3 + q_2q_4) \\ 2(q_1q_2 + q_3q_4) & q_4^2 - q_1^2 + q_2^2 - q_3^2 & 2(q_2q_3 - q_1q_4) \\ 2(q_1q_3 - q_2q_4) & 2(q_2q_3 + q_1q_4) & q_4^2 - q_1^2 - q_2^2 + q_3^2 \end{bmatrix}, \quad (3)$$

where $\mathbf{q}$ describes the rotation from the I -frame to the B -frame. The time derivative of $\mathbf{q}$ is given by

$$\dot{\mathbf{q}} = \frac{1}{2} \begin{bmatrix} q_4 & -q_3 & q_2 & q_1 \\ q_3 & q_4 & -q_1 & q_2 \\ -q_2 & q_1 & q_4 & q_3 \\ -q_1 & -q_2 & -q_3 & q_4 \end{bmatrix} \begin{bmatrix} \omega \\ 0 \end{bmatrix} = \mathbf{Q}(\mathbf{q}) \begin{bmatrix} \omega \\ 0 \end{bmatrix}, \quad (4)$$

where $\omega = [\omega_x \ \omega_y \ \omega_z]'$ is the relative angular velocity of the B -frame with respect to the I -frame given in the B -frame. If the I -frame is inertially fixed, ω is the absolute angular velocity of the B -frame.

B. Attitude parametrization with two quaternions

In order to describe the attitude of the quadrotor helicopter, we parametrize the rotation from an inertial coordinate frame I to a body-fixed coordinate frame B by the quaternion $\mathbf{q}$. The I -frame is a North-East-Down system and the B -frame has its origin in the quadrotor's center of gravity. The essential step for our attitude parametrization is, that we split this rotation described by $\mathbf{q}$ into two linked rotations, which are parametrized by the quaternions $\mathbf{q}_{xy}$ and $\mathbf{q}_z$ respectively. The quaternion $\mathbf{q}_{xy}$ describes the rotation from the I -frame to an auxiliary frame A . The quaternion $\mathbf{q}_z$ describes the rotation from this A -frame to the body-fixed B -frame. The quaternion $\mathbf{q}$, that describes the complete rotation from I to B , reads

$$\mathbf{q} = \mathbf{q}_{xy} \circ \mathbf{q}_z = \mathbf{Q}(\mathbf{q}_{xy}) \cdot \mathbf{q}_z. \quad (5)$$

The auxiliary frame is chosen such that its z -axis is aligned with the z -axis of the B -frame, see Fig. 2. Thus the rotation from the I -frame to the A -frame describes the tilt of the body-fixed z -axis and thus the total thrust direction with respect to the I -frame. As a result, the parameters of $\mathbf{q}_{xy}$

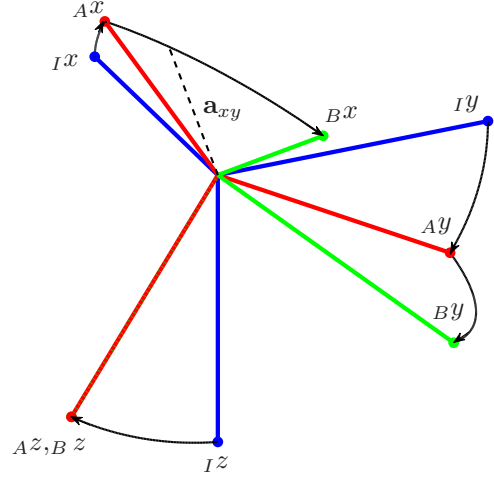


Fig. 2. Rotation from the inertial frame I to the auxiliary frame A and further to the body-fixed frame B .

sufficiently describe the orientation of the total thrust vector. Furthermore, this rotation is about an axis $\mathbf{a}_{xy}$ in the inertial xy -plane, which gives $\mathbf{q}_{xy}$ the following structure

$$\mathbf{q}_{xy} = [q_x \ q_y \ 0 \ q_p]' \quad (6)$$

Since $\mathbf{q}_{xy}$ is a unit quaternion,

$$q_p^2 + q_x^2 + q_y^2 = 1 \quad (7)$$

holds and reveals that $\mathbf{q}_{xy}$ has two degrees of freedom. The remaining rotation from the A -frame to the B -frame is about the body-fixed z -axis and gives $\mathbf{q}_z$ the structure

$$\mathbf{q}_z = [0 \ 0 \ q_z \ q_w]' \quad (8)$$

Again

$$q_w^2 + q_z^2 = 1 \quad (9)$$

holds and shows that $\mathbf{q}_z$ has one degree of freedom. Using (5), we write $\mathbf{q}$ by means of $\mathbf{q}_{xy}$ and $\mathbf{q}_z$ as

$$\mathbf{q} = [q_wq_x + q_zq_y \ q_wq_y - q_zq_x \ q_zq_p \ q_wq_p]' \quad (10)$$

For the elements of $\mathbf{q}_{xy}$, we verify

$$q_p = \sqrt{q_3^2 + q_4^2}, \quad (11)$$

$$q_x = \frac{q_4}{\sqrt{q_3^2 + q_4^2}} \cdot q_1 - \frac{q_3}{\sqrt{q_3^2 + q_4^2}} \cdot q_2, \quad (12)$$

$$q_y = \frac{q_4}{\sqrt{q_3^2 + q_4^2}} \cdot q_2 + \frac{q_3}{\sqrt{q_3^2 + q_4^2}} \cdot q_1, \quad (13)$$

and for the elements of $\mathbf{q}_z$, we use a modified signum function, which has the property $\overline{\text{sgn}}(0) = +1$, to verify

$$q_w = \sqrt{\frac{q_4^2}{q_3^2 + q_4^2}}, \quad q_z = \overline{\text{sgn}}(q_3 \cdot q_4) \cdot \sqrt{\frac{q_3^2}{q_3^2 + q_4^2}}. \quad (14)$$

Thus we have $q_p \geq 0$ and $q_w \geq 0$. This choice eliminates the ambiguity in both quaternions and allows us to design a control law, that prevents unwanted rotations through large angles, which is known as the unwinding phenomenon [12]. Thereby, the $\mathbf{q}_{xy}$ quaternion remains without discontinuities

as long as $q_p > 0$. This is fulfilled as long as the quadrotor is not completely turned up side down. We will choose the desired trajectory such that this is ensured despite moderate tracking errors of the closed loop system. The $\mathbf{q}_z$ quaternion, however, has a discontinuity, which can be understood as a jump in the heading angle from $-\pi$ to π or vice versa. We show, that this does not deteriorate the control performance nor its stability.

C. Parametrization of the thrust vector

With the proposed decomposition of $\mathbf{q}$, it is possible to describe the total thrust vector, which points along the negative z -axis of the B -frame, solely with the $\mathbf{q}_{xy}$ quaternion. The total thrust in the I -frame is given by

$$\mathbf{T} = \mathbf{R}(\mathbf{q}_{xy}) \begin{bmatrix} 0 \\ 0 \\ -T \end{bmatrix} = \begin{bmatrix} -2q_y q_p \\ 2q_x q_p \\ 1 - 2q_p^2 \end{bmatrix} T, \quad (15)$$

where T is the magnitude of the thrust vector. Since $\mathbf{q}_{xy}$ has two degrees of freedom, not only the three components of $\mathbf{q}_{xy}$, but also the time derivatives of these components are indirectly coupled. This can be verified with the derivative of (7), that is

$$\dot{q}_p q_p + \dot{q}_x q_x + \dot{q}_y q_y = 0. \quad (16)$$

With this information, we can derive the time derivative of the thrust vector, which is

$$\dot{\mathbf{T}} = \frac{1}{q_p} \begin{bmatrix} 2Tq_x q_y & 2T(q_y^2 - q_p^2) & -2q_p^2 q_y \\ 2T(q_p^2 - q_x^2) & -2Tq_x q_y & 2q_p^2 q_x \\ 4Tq_x q_p & 4Tq_y q_p & q_p - 2q_p^3 \end{bmatrix} \begin{bmatrix} \dot{q}_x \\ \dot{q}_y \\ \dot{T} \end{bmatrix}. \quad (17)$$

D. Kinematics of the $\mathbf{q}_{xy}$ quaternion

The time derivative of $\mathbf{q}_{xy}$ is according to (4)

$$\dot{\mathbf{q}}_{xy} = \frac{1}{2} \mathbf{Q}(\mathbf{q}_{xy}) \begin{bmatrix} \tilde{\omega} \\ 0 \end{bmatrix}, \quad (18)$$

where $\tilde{\omega}$ is the absolute angular velocity of the A -frame given in the A -frame. Using the fact that the third entry of $\mathbf{q}_{xy}$ is constantly zero, the third element of $\tilde{\omega}$ can be written as

$$\tilde{\omega}_z = \frac{q_y \tilde{\omega}_x - q_x \tilde{\omega}_y}{q_p}. \quad (19)$$

We insert (19) into (18) and obtain

$$\begin{bmatrix} \dot{q}_x \\ \dot{q}_y \end{bmatrix} = \frac{1}{q_p} \cdot \frac{1}{2} \begin{bmatrix} 1 - q_x^2 & -q_x q_y \\ -q_x q_y & 1 - q_y^2 \end{bmatrix} \begin{bmatrix} \tilde{\omega}_x \\ \tilde{\omega}_y \end{bmatrix}. \quad (20)$$

For the time derivative of the thrust vector, we may write (17) with (20) and some reshaping as

$$\dot{\mathbf{T}} = \underbrace{\begin{bmatrix} 2q_x q_y & 2q_y^2 - 1 & -2q_p q_y \\ 1 - 2q_x^2 & -2q_x q_y & 2q_p q_x \\ 2q_x q_p & 2q_y q_p & 1 - 2q_p^2 \end{bmatrix}}_{\mathbf{M}(\mathbf{q}_{xy})} \mathbf{D}(T) \begin{bmatrix} \tilde{\omega}_x \\ \tilde{\omega}_y \\ \dot{T} \end{bmatrix}, \quad (21)$$

where $\mathbf{D}(T) = \text{diag}(T, T, 1)$. Note that $\mathbf{M}(\mathbf{q}_{xy})$ is an orthogonal matrix and thus $\mathbf{M}(\mathbf{q}_{xy})^{-1} = \mathbf{M}(\mathbf{q}_{xy})'$ holds.

E. Kinematics of the $\mathbf{q}_z$ quaternion

Since the z -axis of the A -frame is identical to the z -axis of the B -frame, both frames have the same angular velocity in their common xy -plane. Thus $\tilde{\omega}_{xy} = [\tilde{\omega}_x \ \tilde{\omega}_y]'$ and $\omega_{xy} = [\omega_x \ \omega_y]'$ represent the same angular velocity, but are given in different frames, and $\tilde{\omega}_{xy}$ can be expressed with ω_{xy} by

$$\tilde{\omega}_{xy} = \begin{bmatrix} q_w^2 - q_z^2 & -2q_z q_w \\ 2q_z q_w & q_w^2 - q_z^2 \end{bmatrix} \omega_{xy} = \mathbf{R}_s(\mathbf{q}_z) \omega_{xy}. \quad (22)$$

$\mathbf{R}_s(\mathbf{q}_z)$ is the corresponding submatrix of the transformation matrix $\mathbf{R}(\mathbf{q}_z)$. The angular velocity components in the z -direction are not identical and it is therefore a suitable choice to define the heading angle $\varphi = 2 \arcsin q_z$, which has the time derivative $\dot{\varphi} = \omega_z - \tilde{\omega}_z$. With (19) and (22), we have

$$\tilde{\omega}_z = \frac{1}{q_p} [q_y \ -q_x] \tilde{\omega}_{xy} = \mathbf{l}(\mathbf{q}_{xy})' \mathbf{R}_s(\mathbf{q}_z) \omega_{xy} \quad (23)$$

and write furthermore

$$\dot{\varphi} = \omega_z - \mathbf{l}(\mathbf{q}_{xy})' \mathbf{R}_s(\mathbf{q}_z) \omega_{xy}. \quad (24)$$

F. Dynamic equations of motion

We formulate the quadrotor's dynamic equations used for the control design using the attitude parametrization derived above. For modeling purposes we treat the quadrotor as a rigid body and neglect the rotors' gyroscopic effect, since it has minor influence on the dynamic behavior. The rotation speed of the four rotors is usually controlled by embedded motor controllers, which offer very fast dynamics. Therefore, most control designs, as well as our design, neglect these dynamics and take the four rotor speeds, or equivalently the three torque components, and one total thrust acting on the center of gravity as plant inputs.

The quadrotor's dynamics comprise the following translational dynamics given with respect to the I -frame by

$$\dot{\mathbf{x}} = \mathbf{v}, \quad (25)$$

$$\dot{\mathbf{v}} = \mathbf{f}(\mathbf{v}) + \mathbf{g} + \frac{1}{m} \mathbf{T}, \quad (26)$$

where $\mathbf{x} = [x \ y \ z]'$ is the position, $\mathbf{v} = [\dot{x} \ \dot{y} \ \dot{z}]'$ the velocity, $\mathbf{g} = [0 \ 0 \ g]'$ the gravity vector and $\mathbf{T}$ the total thrust vector. The function $\mathbf{f}(\mathbf{v})$ is twice continuous differentiable and models drag forces. In order to control the translational movement of the quadrotor, we can use ω_{xy} and $\dot{T}$ to change the thrust vector $\mathbf{T}$. The corresponding kinematic relation is given by (22) inserted into (21), which is

$$\dot{\mathbf{T}} = \mathbf{M}(\mathbf{q}_{xy}) \mathbf{D}(T) \mathbf{R}(\mathbf{q}_z) [\omega'_{xy} \ \dot{T}]'. \quad (27)$$

The rotation ω_z has no influence on the translational dynamics and can independently be used to control the heading angle according to (24). To this end, we use the Euler's equations, with $\mathbf{J}$ being the moment of inertia,

$$\dot{\omega} = \mathbf{J}^{-1} (\mathbf{J} \omega \times \omega + \tau), \quad (28)$$

to access both, $\dot{\omega}_z$ for the heading angle control and ω_{xy} for the position tracking. Thus we choose the three torque

components $\boldsymbol{\tau} = [\tau_x \ \tau_y \ \tau_z]'$ as control inputs. For the control design, we want to use the input transformation

$$\boldsymbol{\tau} = -\mathbf{J}\boldsymbol{\omega} \times \boldsymbol{\omega} + \mathbf{J} \tilde{\boldsymbol{\tau}}, \quad (29)$$

which gives us direct access to the angular acceleration

$$\dot{\boldsymbol{\omega}} = \tilde{\boldsymbol{\tau}}. \quad (30)$$

Referring to (27), the translational movement of the quadrotor is also influenced by $\dot{T}$ and consequently by the second time derivative $\ddot{T}$. Without a dynamic plant extension for the thrust magnitude, one can not access the derivatives $\dot{T}$ and $\ddot{T}$. Thus many contributions, e.g. [7], [13], use a dynamic plant extension for the thrust command. Applying the backstepping approach, one can avoid such a dynamic extension and realize the desired thrust directly by the actuators [8], [14]. Even though this may yield a tighter height control, the dynamic plant extension is still useful to obtain a smooth and less aggressive behavior. Hence we extend the model by two integrators for the thrust magnitude and choose

$$\ddot{T} = u_T \quad (31)$$

as further control input.

III. BACKSTEPPING TRAJECTORY TRACKING CONTROL

We develop the backstepping trajectory tracking controller with the model derived above. We design the position tracking control in the first part and the heading tracking control in the second part. For ease of presentation, we omit the arguments of $\mathbf{M}(\mathbf{q}_{xy})$, $\mathbf{D}(T)$, $\mathbf{R}(\mathbf{q}_x)$ and $\mathbf{l}(\mathbf{q}_{xy})$ in this section.

A. Position tracking control

As a first step, we define the position tracking error $\mathbf{z}_x = \mathbf{x} - \mathbf{x}_T$, with $\mathbf{x}_T$ being the desired trajectory, and choose the Lyapunov function $V_x = \frac{1}{2} \mathbf{z}_x' \mathbf{z}_x$ to asymptotically stabilize the origin $\mathbf{z}_x = 0$. The time derivative of V_x is

$$\dot{V}_x = \mathbf{z}_x' (\mathbf{v} - \dot{\mathbf{x}}_T), \quad (32)$$

which leads to the desired velocity

$$\mathbf{v}_d = \dot{\mathbf{x}}_T + \mathbf{A}_x \mathbf{z}_x, \quad (33)$$

with $\mathbf{A}_x < 0$, that drives $V_x \rightarrow 0$. As long as the velocity $\mathbf{v} \neq \mathbf{v}_d$, we have $\mathbf{v} - \dot{\mathbf{x}}_T = \mathbf{z}_v + \mathbf{A}_x \mathbf{z}_x$, with the velocity error defined as $\mathbf{z}_v = \mathbf{v} - \mathbf{v}_d$, and (32) actually reads

$$\dot{V}_x = \mathbf{z}_x' \mathbf{A}_x \mathbf{z}_x + \mathbf{z}_x' \mathbf{z}_v. \quad (34)$$

Consequently, we want to drive $\mathbf{z}_v \rightarrow 0$ and therefore we extend V_x in the following way

$$V_v = V_x + \frac{1}{2} \mathbf{z}_v' \mathbf{z}_v \quad (35)$$

and obtain

$$\dot{V}_v = \mathbf{z}_x' \mathbf{A}_x \mathbf{z}_x + \mathbf{z}_v' (\mathbf{z}_x + \dot{\mathbf{v}} - \dot{\mathbf{v}}_d). \quad (36)$$

According to (26), $\dot{\mathbf{v}}$ can be influenced by the total thrust $\mathbf{T}$. Thus the desired thrust vector, that drives V_v to zero, is

$$\mathbf{T}_d = m(-\mathbf{f}(\mathbf{v}) - \mathbf{g} + \dot{\mathbf{v}}_d - \mathbf{z}_x + \mathbf{A}_v \mathbf{z}_v), \quad (37)$$

with $\mathbf{A}_v < 0$ and $\dot{\mathbf{v}}_d = \ddot{\mathbf{x}}_T + \mathbf{A}_x(\mathbf{v} - \dot{\mathbf{x}}_T)$.

In the same manner as previously, as long as the thrust vector $\mathbf{T} \neq \mathbf{T}_d$, we have $\dot{\mathbf{v}} - \dot{\mathbf{v}}_d = \frac{1}{m}(\mathbf{T} - \mathbf{T}_d) - \mathbf{z}_x + \mathbf{A}_v \mathbf{z}_v$. With the thrust error defined as $\mathbf{z}_t = \mathbf{T} - \mathbf{T}_d$, we rewrite (36) as

$$\dot{V}_v = \mathbf{z}_x' \mathbf{A}_x \mathbf{z}_x + \mathbf{z}_v' \mathbf{A}_v \mathbf{z}_v + \frac{1}{m} \mathbf{z}_v' \mathbf{z}_t. \quad (38)$$

In the next step, we want to drive $\mathbf{z}_t \rightarrow 0$ and extend V_v as follows

$$V_t = V_v + \frac{1}{2} \mathbf{z}_t' \mathbf{z}_t. \quad (39)$$

Therein, $\mathbf{z}_t' \mathbf{z}_t$ can be written with the smallest angle α_e between $\mathbf{T}$ and $\mathbf{T}_d$ and the magnitude error $\tilde{T} = T - T_d$ as $\mathbf{z}_t' \mathbf{z}_t = \tilde{T}^2 + 2T_d(\tilde{T} + T_d)(1 - \cos \alpha_e)$. Thus our error compensation derived below seeks both, to eliminate the magnitude error $\tilde{T}$ and to rotate through the shortest path to eliminate the angle error α_e .

In order to achieve this, we derive $\dot{V}_t$, which is

$$\dot{V}_t = \mathbf{z}_x' \mathbf{A}_x \mathbf{z}_x + \mathbf{z}_v' \mathbf{A}_v \mathbf{z}_v + \mathbf{z}_t' \left(\frac{1}{m} \mathbf{z}_v + \dot{\mathbf{T}} - \dot{\mathbf{T}}_d \right). \quad (40)$$

As given in (27), $\dot{\mathbf{T}}$ may be changed through a variation of ω_{xy} and $\dot{T}$. Since we want to drive $V_t \rightarrow 0$, we assign, with $\mathbf{A}_t < 0$,

$$\begin{bmatrix} \omega_{xyd} \\ \dot{T}_d \end{bmatrix} = \mathbf{R}' \mathbf{D}^{-1} \mathbf{M}' \left(\dot{\mathbf{T}}_d + \mathbf{A}_t \mathbf{z}_t - \frac{1}{m} \mathbf{z}_v \right). \quad (41)$$

Note that the inverse matrix $\mathbf{D}^{-1} = \text{diag}(\frac{1}{T}, \frac{1}{T}, 1)$ requires $T \neq 0$. This is a common and system-inherent obstacle all position tracking control algorithms of quadrotors have to deal with. To remedy this problem, we will shape the desired trajectory so that $T > 0$ is ensured with enough reserve to cope with reasonable disturbances.

The time derivative of $\mathbf{T}_d$ can easily be obtained by deriving (37), which is $\dot{\mathbf{T}}_d = m(-\dot{\mathbf{f}} + \ddot{\mathbf{x}}_T + \mathbf{A}_x(\dot{\mathbf{v}} - \dot{\mathbf{x}}_T) - (\mathbf{v} - \dot{\mathbf{x}}_T) - \mathbf{A}_v(\dot{\mathbf{v}} - \dot{\mathbf{v}}_d))$. The acceleration $\dot{\mathbf{v}}$ therein is calculated using (26).

Similar to the previous steps, we have $\dot{\mathbf{T}} - \dot{\mathbf{T}}_d = \mathbf{MDR} \mathbf{z}_r + \mathbf{A}_t \mathbf{z}_t - \frac{1}{m} \mathbf{z}_v$ with the angular velocity error $\mathbf{z}_r = [\omega'_{xy} - \omega'_{xyd} \quad \dot{T} - \dot{T}_d]'$. We combine that with (40) and obtain

$$\dot{V}_t = \mathbf{z}_x' \mathbf{A}_x \mathbf{z}_x + \mathbf{z}_v' \mathbf{A}_v \mathbf{z}_v + \mathbf{z}_t' \mathbf{A}_t \mathbf{z}_t + \mathbf{z}_t' \mathbf{MDR} \mathbf{z}_r. \quad (42)$$

In the next step, we extend V_t once more in such a way that

$$V_\omega = V_t + \frac{1}{2} \mathbf{z}_r' \mathbf{z}_r, \quad (43)$$

in order to achieve $\mathbf{z}_r \rightarrow 0$. The time derivative is

$$\dot{V}_\omega = \mathbf{z}_x' \mathbf{A}_x \mathbf{z}_x + \mathbf{z}_v' \mathbf{A}_v \mathbf{z}_v + \mathbf{z}_t' \mathbf{A}_t \mathbf{z}_t + \mathbf{z}_r' \left(\mathbf{R}' \mathbf{D} \mathbf{M}' \mathbf{z}_t + [\dot{\omega}'_{xy} - \dot{\omega}'_{xyd} \quad \ddot{T} - \ddot{T}_d]' \right). \quad (44)$$

Now, we can use u_T in (31) and the first two components of (30), which are $\dot{\omega}_{xy} = \tilde{\tau}_{xy}$, as control inputs. For these control inputs, we choose

$$\begin{bmatrix} \tilde{\tau}'_{xy} & u_T \end{bmatrix}' = [\dot{\omega}'_{xyd} \quad \ddot{T}_d]' + \mathbf{A}_r \mathbf{z}_r - \mathbf{R}' \mathbf{D} \mathbf{M}' \mathbf{z}_t, \quad (45)$$

with $\mathbf{A}_r < 0$, and render (44) negative definite, which is

$$\dot{V}_\omega = \mathbf{z}'_x \mathbf{A}_x \mathbf{z}_x + \mathbf{z}'_v \mathbf{A}_v \mathbf{z}_v + \mathbf{z}'_t \mathbf{A}_t \mathbf{z}_t + \mathbf{z}'_r \mathbf{A}_r \mathbf{z}_r. \quad (46)$$

Hence (45) is an asymptotically stabilizing control law for the position trajectory tracking problem. The dynamics of the heading angle, given in (24), have no influence on this stability result. Furthermore, we can provide the desired derivatives in (45) with ease. These are $[\dot{\omega}'_{xyd} \quad \ddot{T}_d]' = \frac{d}{dt} (\mathbf{R}' \mathbf{D}^{-1} \mathbf{M}') (\dot{\mathbf{T}}_d + \mathbf{A}_t \mathbf{z}_t - \frac{1}{m} \mathbf{z}_v) + \mathbf{R}' \mathbf{D}^{-1} \mathbf{M}' (\dot{\mathbf{T}}_d + \mathbf{A}_t \dot{\mathbf{z}}_t - \frac{1}{m} \dot{\mathbf{z}}_v)$, where $\ddot{\mathbf{T}}_d = m(-\dot{\mathbf{f}} + \ddot{\mathbf{x}}_T + \mathbf{A}_x(\ddot{\mathbf{v}} - \ddot{\mathbf{x}}_T) - (\dot{\mathbf{v}} - \dot{\mathbf{x}}_T) - \mathbf{A}_v(\ddot{\mathbf{v}} - \ddot{\mathbf{v}}_d))$ and $\ddot{\mathbf{v}} = \dot{\mathbf{f}} + \frac{1}{m} \dot{\mathbf{T}}_d$. The calculation of $\dot{\mathbf{R}}'$, $\mathbf{D}^{-1}$ and $\dot{\mathbf{M}}'$ is straight forward and omitted due to limited space.

B. Heading angle tracking control

For the asymptotic stabilization of the heading angle tracking, we choose the Lyapunov function

$$V_\varphi = \frac{1}{2} z_\varphi^2. \quad (47)$$

Therein, $z_\varphi = \text{mod}(\varphi - \varphi_T + \pi, 2\pi) - \pi$ defines the heading tracking error, with the modulo function $\text{mod}(x, y)$ and the desired time varying heading angle $\varphi_T \in [-\pi, \pi]$. This definition renders $z_\varphi \in [-\pi, \pi]$. It further prevents an unwanted change in direction of the heading course, in case φ_T or φ performs a jump from $-\pi$ to π or vice versa. However, due to a disturbance it may happen that z_φ jumps from $-\pi$ to π or vice versa. We will show later on that when we hit this discontinuity the asymptotic stability of the closed loop system is still ensured.

The time derivative of (47) is $\dot{V}_\varphi = z_\varphi \dot{z}_\varphi = z_\varphi (\dot{\varphi} - \dot{\varphi}_T)$, with a discontinuity that appears when z_φ performs a jump. To drive $z_\varphi \rightarrow 0$, we choose, using (24), the following desired angular velocity in z -direction

$$\omega_{zd} = \dot{\varphi}_T + a_\varphi z_\varphi + \mathbf{l}' \mathbf{R}_s \boldsymbol{\omega}_{xy}, \quad (48)$$

with $a_\varphi < 0$. Similar to the steps in the previous section, we define the angular velocity error $z_{\omega z} = \omega_z - \omega_{zd}$ and rewrite $\dot{V}_\varphi$ with $\dot{z}_\varphi = a_\varphi z_\varphi + z_{\omega z}$ as

$$\dot{V}_\varphi = a_\varphi z_\varphi^2 + z_\varphi z_{\omega z}. \quad (49)$$

In order to drive $z_{\omega z} \rightarrow 0$, we extent (47) to be

$$V_{\omega z} = V_\varphi + \frac{1}{2} z_{\omega z}^2. \quad (50)$$

The time derivative thereof is

$$\dot{V}_{\omega z} = a_\varphi z_\varphi^2 + z_{\omega z} (z_\varphi + \dot{\omega}_z - \dot{\omega}_{zd}), \quad (51)$$

which leads to the choice

$$\tilde{\tau}_z = \dot{\omega}_z = \dot{\omega}_{zd} + a_{\omega z} z_{\omega z} - z_\varphi, \quad (52)$$

with $a_{\omega z} < 0$, to render $\dot{V}_{\omega z} = a_\varphi z_\varphi^2 + a_{\omega z} z_{\omega z}^2$ negative definite. We calculate the derivative $\dot{\omega}_{zd}$ with a derivation of (23) and with $\dot{\omega}_{xy} = \tilde{\tau}_{xy}$ and obtain $\dot{\omega}_z = \dot{\varphi}_T + a_\varphi z_\varphi + (\mathbf{l}' \mathbf{R}_s + \mathbf{l}' \dot{\mathbf{R}}_s) \boldsymbol{\omega}_{xy} + \mathbf{l}' \mathbf{R}_s \tilde{\tau}_{xy}$. The calculation of $\dot{\mathbf{l}}$ and $\dot{\mathbf{R}}_s$ is

straight forward and therefore omitted. To sum up, the closed loop dynamics for the heading control is

$$\dot{z}_\varphi = a_\varphi z_\varphi + z_{\omega z} \quad (53)$$

$$\dot{z}_{\omega z} = a_{\omega z} z_{\omega z} - z_\varphi. \quad (54)$$

To show the asymptotic stability of the heading control, we additionally need to examine the discontinuity of z_φ . The discontinuity occurs, when z_φ is on the boundary of $[-\pi, \pi[$ and $\dot{z}_\varphi$ points into the exterior of $[-\pi, \pi[$. With (48), we can rewrite (53) as $\dot{z}_\varphi = \omega_z - \dot{\varphi}_T - \mathbf{l}' \mathbf{R}_s \boldsymbol{\omega}_{xy}$. This reveals that $\dot{z}_\varphi$ is not affected by the discontinuity and retains its direction after a jump of z_φ , which will drive the system away from the discontinuity. This is enough to conclude asymptotic stability of the heading angle tracking control.

The complete control law for the position and heading trajectory tracking problem is given by (45) and (52). The parameters $\mathbf{A}_x$, $\mathbf{A}_v$, $\mathbf{A}_t$, $\mathbf{A}_r$, a_φ and $a_{\omega z}$ are left to be chosen. In this regard our design is very transparent, because each parameter can be associated with the convergence of a characteristic group of state variables. To assign increasing eigenvalues from the position dynamics to the angular velocity dynamics is an intuitive and reasonable choice, since the system has a cascaded structure. Although other choices lead to asymptotic convergence, this choice provided much better experimental results, which are presented in the next section.

IV. EXPERIMENTAL RESULTS

This section shows first results of experimental indoor flights, that we performed with an AscTec Hummingbird (www.ascotec.de). We measured the position, velocity and heading information with a motion capturing system from Vicon (www.vicon.com) and transmitted this data to the quadrotor via radio link operating at 50 Hz. Together with the data from the onboard gyroscope and acceleration sensors, we could reconstruct all state variables onboard. The control algorithm was running on an ARM Cortex-A8 processor at 1 kHz (www.arm.com) and required 16% of the available CPU power, which reveals a rather low computational cost. For the parametrization of the tracking controller, we chose, as discussed in the section above, $\mathbf{A}_x = -\mathbf{I}_{3 \times 3}$, $\mathbf{A}_v = -3 \cdot \mathbf{I}_{3 \times 3}$, $\mathbf{A}_t = -8 \cdot \mathbf{I}_{3 \times 3}$, $\mathbf{A}_r = -12 \cdot \mathbf{I}_{3 \times 3}$ and for the heading controller $a_\varphi = -2$ and $a_{\omega z} = -12$. The intension of these experiments was to provide a proof of concept rather than to optimize the performance. Since we are aiming for an adaptive extension to avoid costly parameter identification, we used a rough estimation for the model parameters in these experiments and evaluate the outcome as intermediate results.

In the test flight shown in Fig. 3, we commanded switching set points for the position and heading angle. The commanded signals, denoted by $\mathbf{x}_C$ and φ_C , served as input signals for a reference model that provided the sufficiently smooth trajectories $\mathbf{x}_T \in C^4$ and $\varphi_T \in C^2$. Although the algorithm has no integral terms, the height and heading trajectory was precisely tracked with small overshoots and almost no delay. In the underactuated horizontal directions, we observe a bit more deviations. During set point transition,

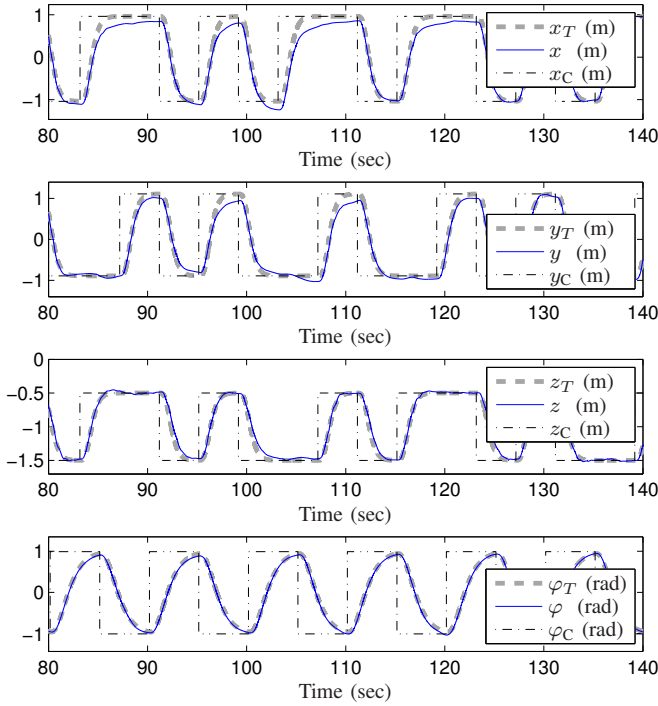


Fig. 3. Transition between switched set points.

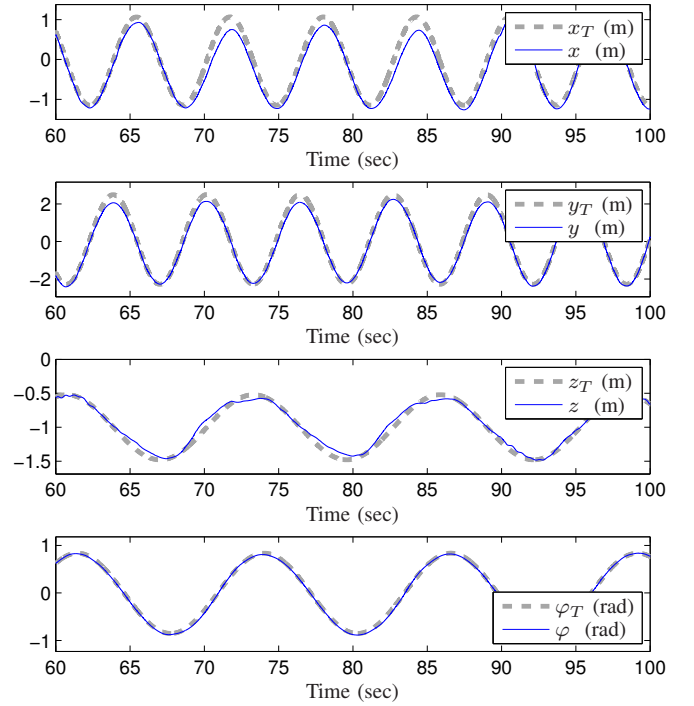


Fig. 4. Tracking of a vertical helix.

the quadrotor had a maximum velocity of 2.2 m/s and the thrust vector was tilted against the upright position with a maximum of 22 degrees.

The tracking of a vertical helix is shown in Fig. 4. The height and heading angle command oscillated with a period time of 12.5 sec. The period time for the horizontal positions was 6.25 sec. During flight, we measured a maximum velocity of 1.4 m/s and the thrust vector was tilted with a maximum of 18 degrees. The quadrotor tracked the trajectory without phase delay and showed a precise heading angle tracking. The achieved amplitude in the translational directions was a bit smaller than desired and since the height trajectory was comparatively slow, we can notice an influence of the horizontal directions on the height tracking.

V. CONCLUSIONS

The decoupling attitude parametrization allowed, on the one hand, an independent and straightforward design of the position tracking control and the heading tracking control. On the other hand, we could analytically derive all control signals with reasonable expense. We could further define and handle a meaningful attitude error in $\mathbb{R}^3$ and thereby keep the equations clear. Moreover, it was possible to choose the parameters for the error dynamics in a transparent manner. The experimental results showed the practical suitability of the suggested control algorithm.

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